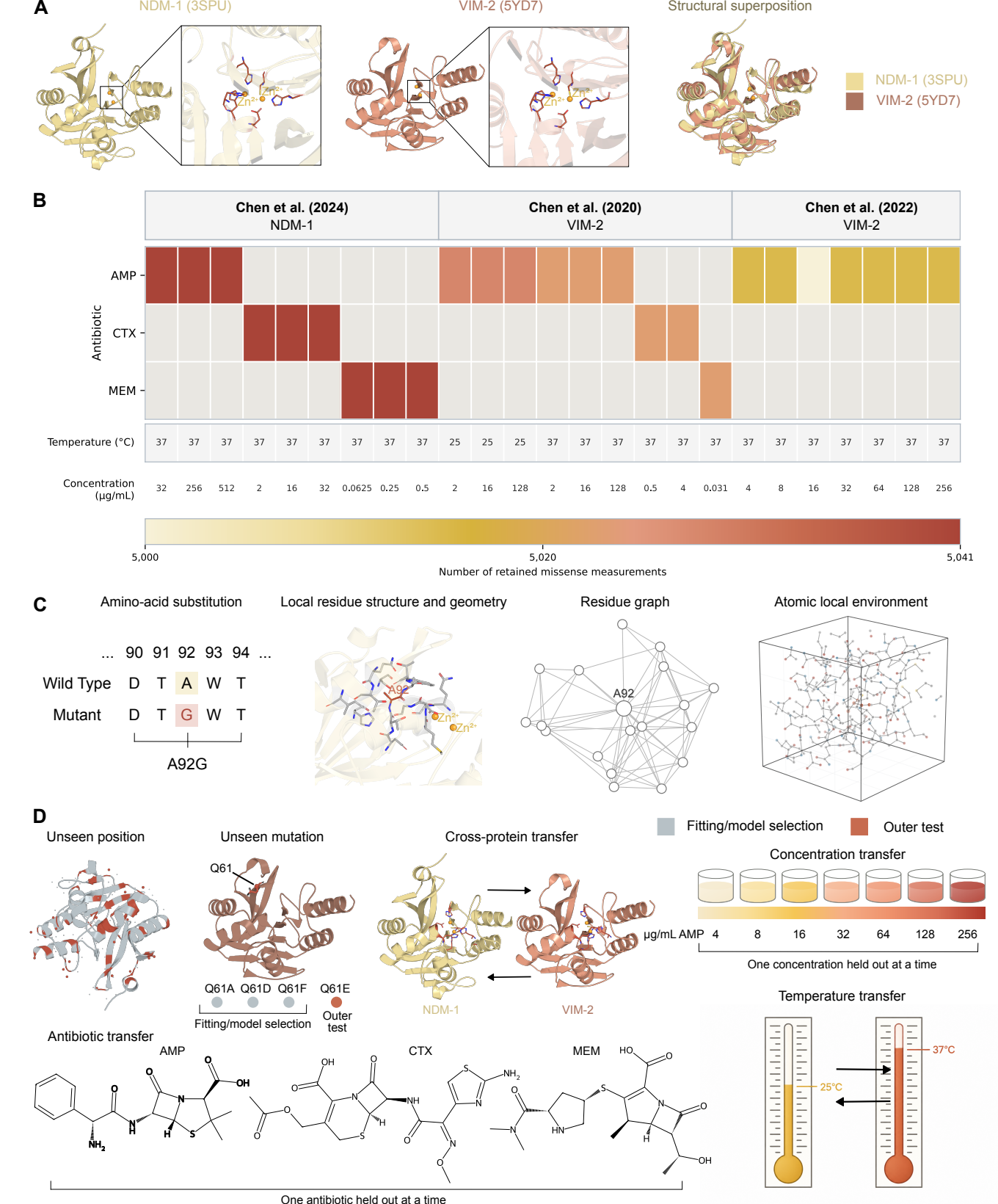

A
NDM-1 (3SPU)
VIM-2 (5YD7)
Structural superposition
NDM-1 (3SPU)
VIM-2 (5YD7)
B
Chen et al. (2024)
NDM-1
Chen et al. (2020)
VIM-2
Chen et al. (2022)
VIM-2
Antibiotic
AMP
CTX
MEM
Temperature (°C)
Concentration (μg/mL)
5,000
5,020
5,041
Number of retained missense measurements
C
Amino-acid substitution
Local residue structure and geometry
Residue graph
Atomic local environment
... 90 91 92 93 94 ...
Wild Type D T A W T
Mutant D T G W T
A92G
A92
D
Unseen position
Unseen mutation
Cross-protein transfer
Fitting/model selection
Outer test
Concentration transfer
μg/mL AMP 4 8 16 32 64 128 256
One concentration held out at a time
Q61
Q61A Q61D Q61F
Q61E
Fitting/model selection
Outer test
NDM-1
VIM-2
Temperature transfer
25°C
37°C
Antibiotic transfer
AMP
CTX
MEM
One antibiotic held out at a time